Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary, Renal
Lecture: CPR 53

Title: Flipped Classroom: Interpreting Acid Base Disorders

Please answer these questions before the lecture

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Flipped Class: Interpreting Acid Base Disorders

SOM.MK.I.BPM1.3.CPR.1.INTG. 0024	Distinguish between acute and chronic respiratory disturbances. Describe the causes, compensatory responses for: Acute and Chronic Respiratory acidosis – Causes of hypoventilation. Acute and Chronic Respiratory alkalosis – Causes of hyperventilation, high altitude adaptations.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0025	Interpret ABG data (pH, PCO2 and $HCO3^-$) and solve problems related to disorders of acid base balance using Davenport diagram. From blood values, identify simple and mixed metabolic and respiratory acid-base disturbances.

Recommended Reading

- Medical Physiology: Principles for Clinical Medicine, R. Rhoades & D. Bell; 6th ed.; Chapter 24, pp. 533-555; Acid–Base Homeostasis | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library (oclc.org)
- Costanzo Physiology; https://www.clinicalkey.com/student/content/book/3-s2.0-B978032379333900016X
- Ref: Mark's Essentials of Biochemistry: Chapter 4; Water, Acids, Bases, and Buffers | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library (oclc.org)
- Reference: Vander's Renal Physiology; Chapter 9: Acid base balance; Regulation of Acid-Base Balance | Vander's Renal Physiology, 10th Edition | AccessMedicine | McGraw Hill Medical (oclc.org)
- Additional problems: http://www.scribd.com/doc/6442924/ABG-Interpretation Slide 17 onwards
- http://www.mhhe.com/biosci/ap/ap_casestudies/cases/ap_case14.html
- Practice questions folder for the week; Small group case

Ready!!!

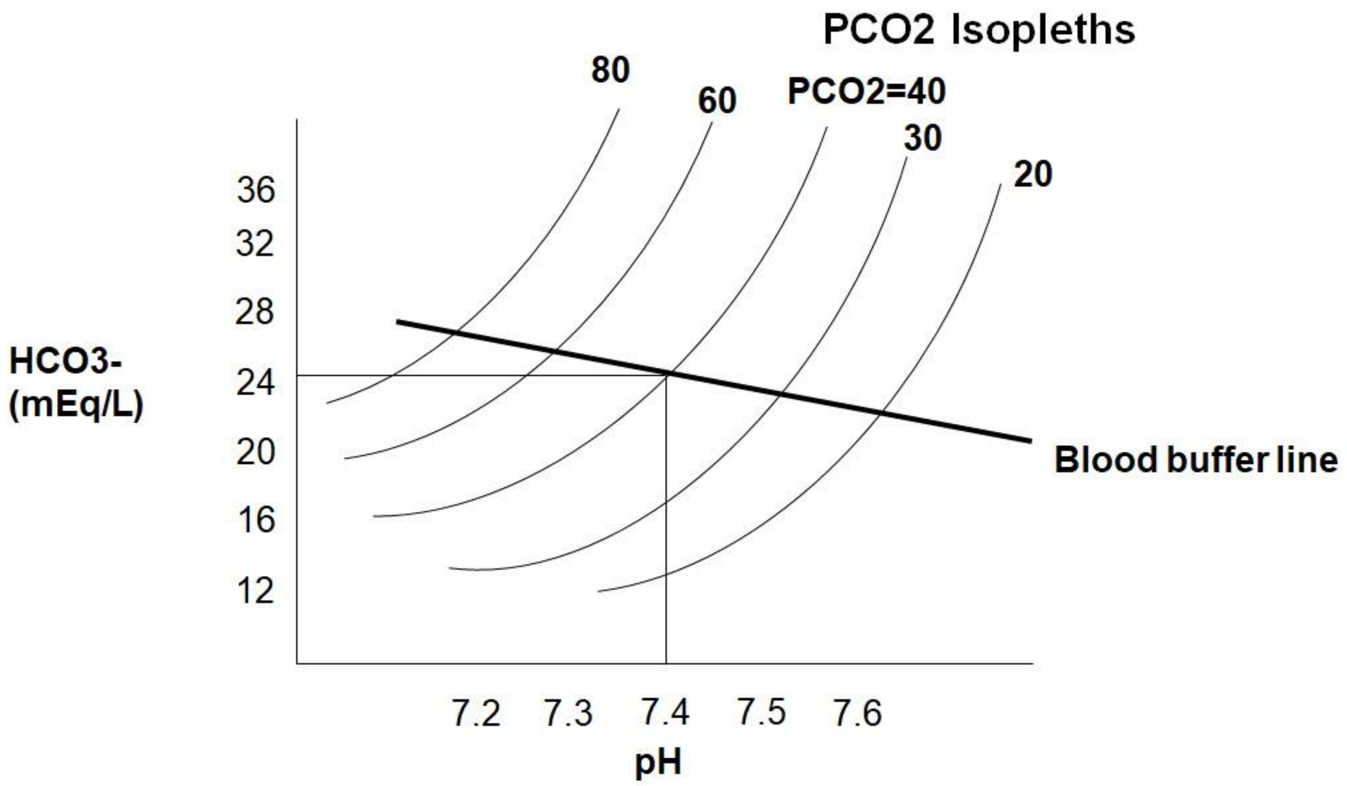
Normal arterial blood gas levels:

pH=7.4 (7.35-7.45)

$PaCO_2$ =40 mmHg (33-45)

HCO_3^- =24 mEq/L (22-28)

Plot on the Davenport diagram: pH = 7.49; PCO_2 = 50mmHg; HCO_3^- = 38mmol/L



Identify the acid base disturbance;
Name two disorders;
Identify compensatory mechanisms

Summary

- First look at pH: Acidosis or Alkalosis
- Next, what is primary problem? Metabolic or Respiratory?
  - Look at PCO_2 and pH, (**ROMS = Respiratory Opposite; Metabolic Same)**
    - If PCO_2 change is opposite direction of pH change, then it is RESPIRATORY
      - Eg: If pH- low and PCO_2- high **(RESPIRATORY)**
    - If PCO_2 change is same direction as pH change, then it is METABOLIC
      - Eg: If pH-low and PCO_2- low **(METABOLIC);** confirm by looking at HCO_3^-
- Check whether Chronic or Acute (Respiratory disorders)
  - Look at HCO_3^-
    - If HCO_3^- values in the same direction as PCO_2 (Chronic respiratory)
    - If HCO_3^- levels within normal limits (Acute respiratory)

A 4-year-old boy is brought to the emergency room by his mother because he has accidentally ingested aspirin tablets 2-hours ago. His respiratory rate is markedly increased. A diagnosis of acute salicylate poisoning is made. What may be the predicted arterial blood results?

- pH =
- $PaCO_2$ =
- HCO_3^- =

Chronic salicylate poisoning results in metabolic acidosis because of formation of weak acids (due to salicylate metabolism). What may be the predicted arterial blood results?

A. pH-↑; PCO_2-low; HCO_3^--low	17%
B. pH-↓; PCO_2-Normal; HCO_3^--low	17%
C. pH-↑; PCO_2-high; HCO_3^--high	17%
D. pH-↓; PCO_2-low; HCO_3^--low	17%
E. pH-↓; PCO_2-high; HCO_3^--Normal	17%
F. pH-↑; PCO_2-low; HCO_3^--Normal	17%

A 45-year-old man has the following arterial blood gas reports: pH =7.13; PCO_2 = 55mmHg; HCO_3^- = 18mEq/L.

- What is the most likely acid-base disturbance?

A 38-year-old woman has the following arterial blood gas reports: pH =7.56; PCO_2 = 33 mmHg; HCO_3^- = 29 mEq/L.

- What is the most likely acid-base disturbance?

Mixed Acid-base Disorders

- Presence of more than one acid-base disorder
- Knowledge of clinical picture essential for interpretation

- **PCO_2 and HCO_3^-** move in **opposite directions** (remember, compensation always moves in the same direction)
- If the compensation is not adequate, based on calculations
- If the pH is normal and HCO_3^- and PCO_2 are abnormal

Mixed Acid-base Disorders

- Examples:
  - Salicylate poisoning – stimulation of respiratory center (respiratory alkalosis)/effect on metabolism (metabolic acidosis)
  - Hyperventilation (respiratory alkalosis) with prolonged nasogastric suction (metabolic alkalosis)
  - Chronic obstructive airways disease (respiratory acidosis) with thiazide diuretic -induced K^+ depletion (metabolic alkalosis)
  - Chronic bronchitis (respiratory acidosis) with renal impairment (metabolic acidosis)

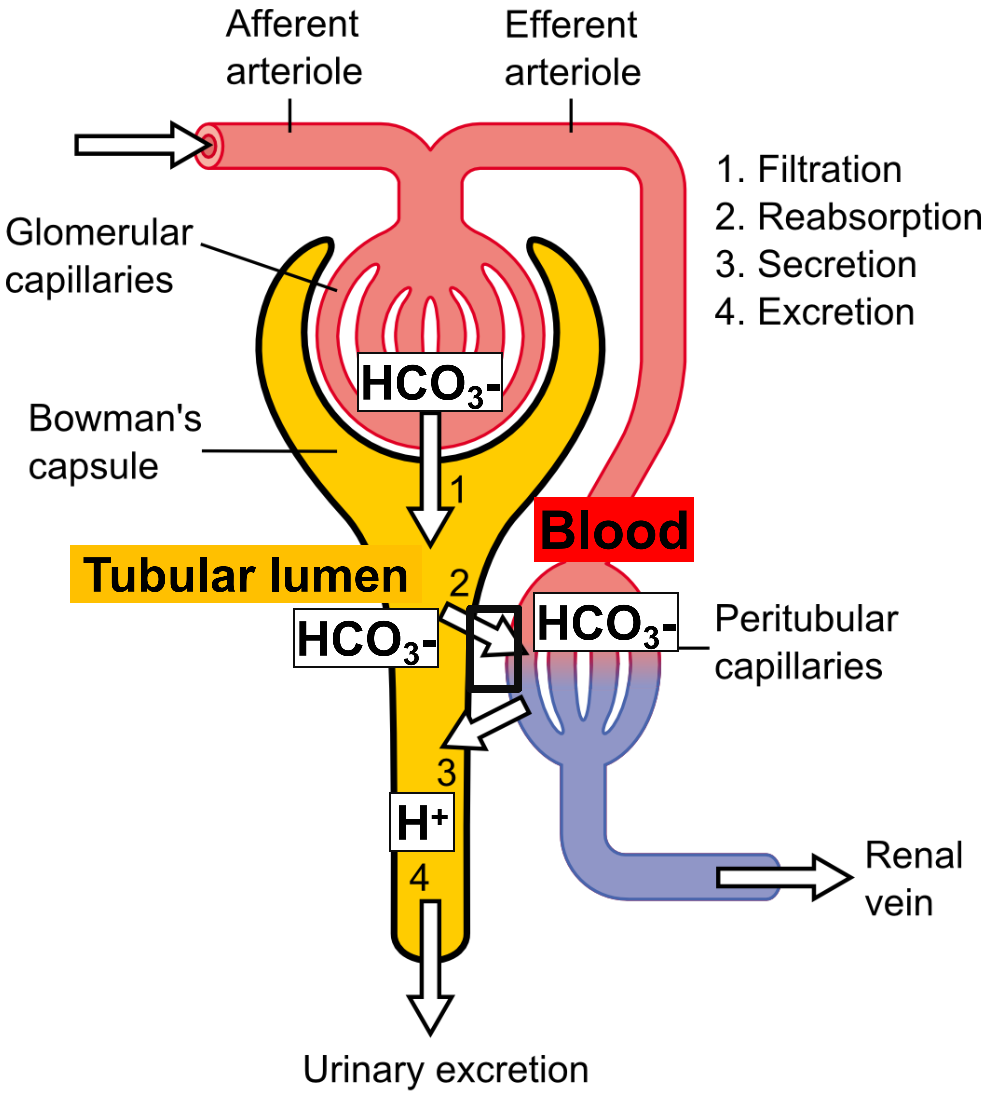
A 60-year-old woman has the following reports: pH = 7.12; HCO_3^- = 24mmol/L; $PaCO_2$ = 75 mmHg; PaO_2=70 mmHg.

- What is the most likely acid base disturbance?
- Why are the $PaCO_2$ and PaO_2 abnormal?
- What compensatory mechanisms will most likely be active in this patient?
- Predict the ABG after 5 days (Assume not treated)
- Name 4 disorders that can result in this acid-base disturbance.

A 35-year-old woman is admitted to the ICU with a 2-day history of difficulty breathing and walking. She had gastroenteritis a few weeks ago, and a diagnosis of Guillain-Barré syndrome is made. No improvement is observed after 4 days. What compensatory mechanisms are most likely active in the patient?

A. Urinary bicarbonate increases	17%
B. Urinary acid phosphate (titratable acid) decreases	17%
C. Urinary ammonium increases	17%
D. Urinary pH increases	17%
E. Urine pH becomes basic	17%
F. Tubular carbonic anhydrase is inhibited	17%

Renal compensation in respiratory/ metabolic acidosis



- In chronic acidosis (if kidney is normal),
- Increased H^+ secretion by tubule
- Filtered bicarbonate is reabsorbed
- Acid phosphate (titratable acidity) is increased
- Ammonium in urine is increased
- Urine pH becomes acidic

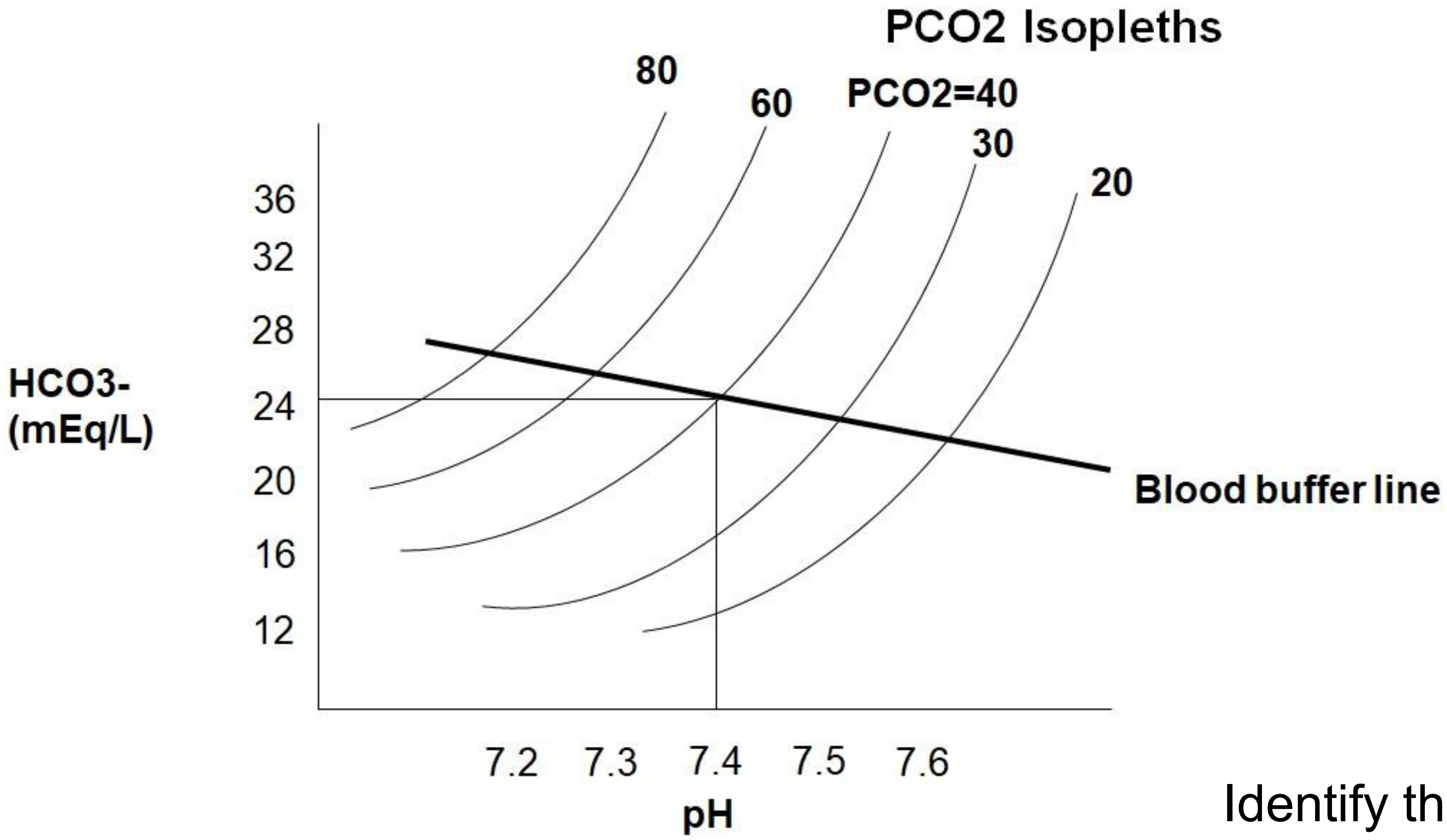
Increased phosphate and ammonium excretion in urine

A 20-year-old man is admitted because of a 4-hour history of accidental intake of methanol. ABG reports show: pH = 7.25; $PaCO_2$ = 30 mmHg; HCO_3^- = 13 mmol/L.

- What is the acid base disturbance?
- Why is the HCO_3^- level low?
- Comment on respiratory rate
- Explain compensatory mechanisms active in this patient?

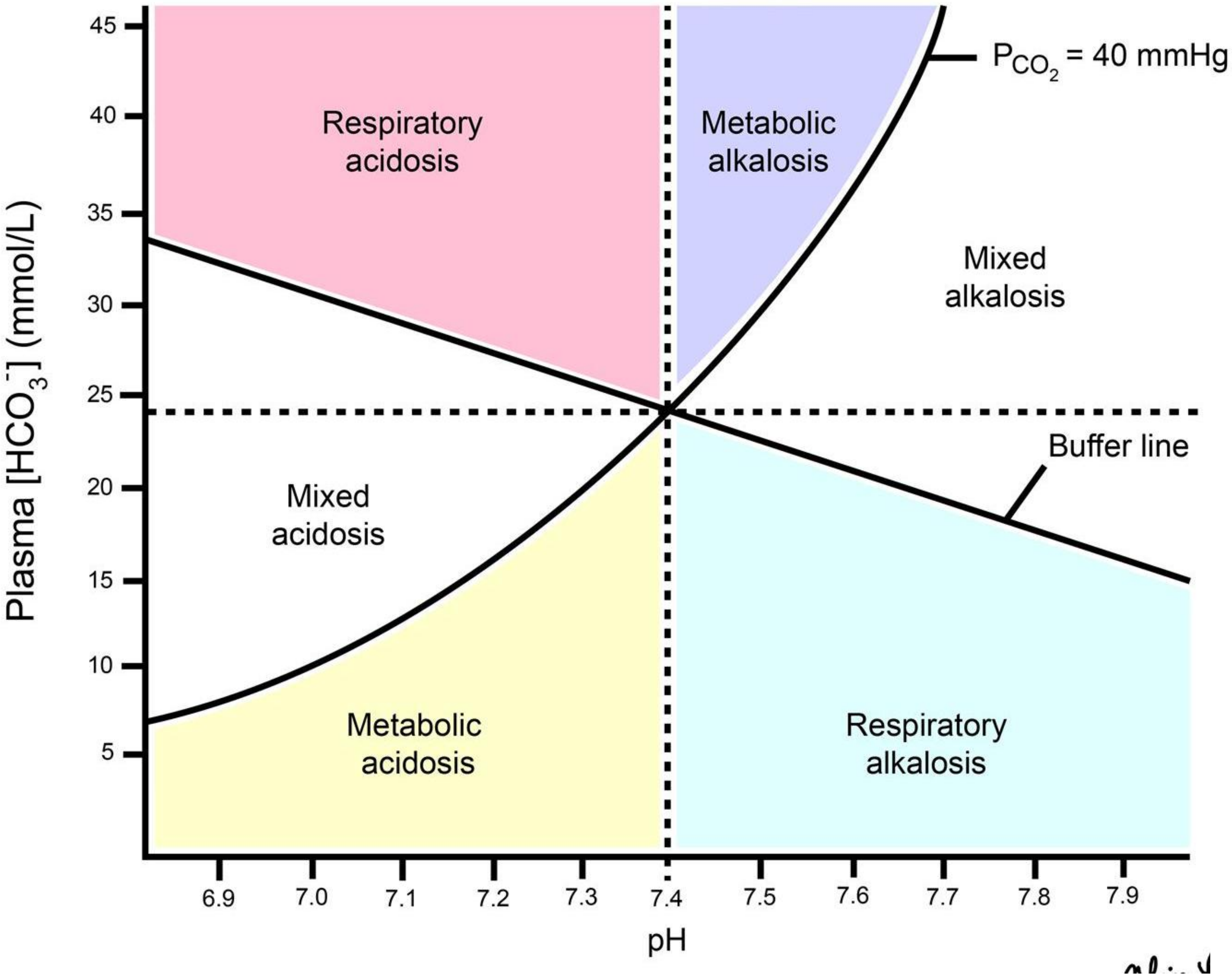
- Name 4 disorders which may result in this acid base disturbance.

A 10-year-old girl with a 1-day history of fever and productive cough shows the following ABG results: pH = 7.23; HCO_3^- = 25mmol/L; PCO_2 = 60mmHg



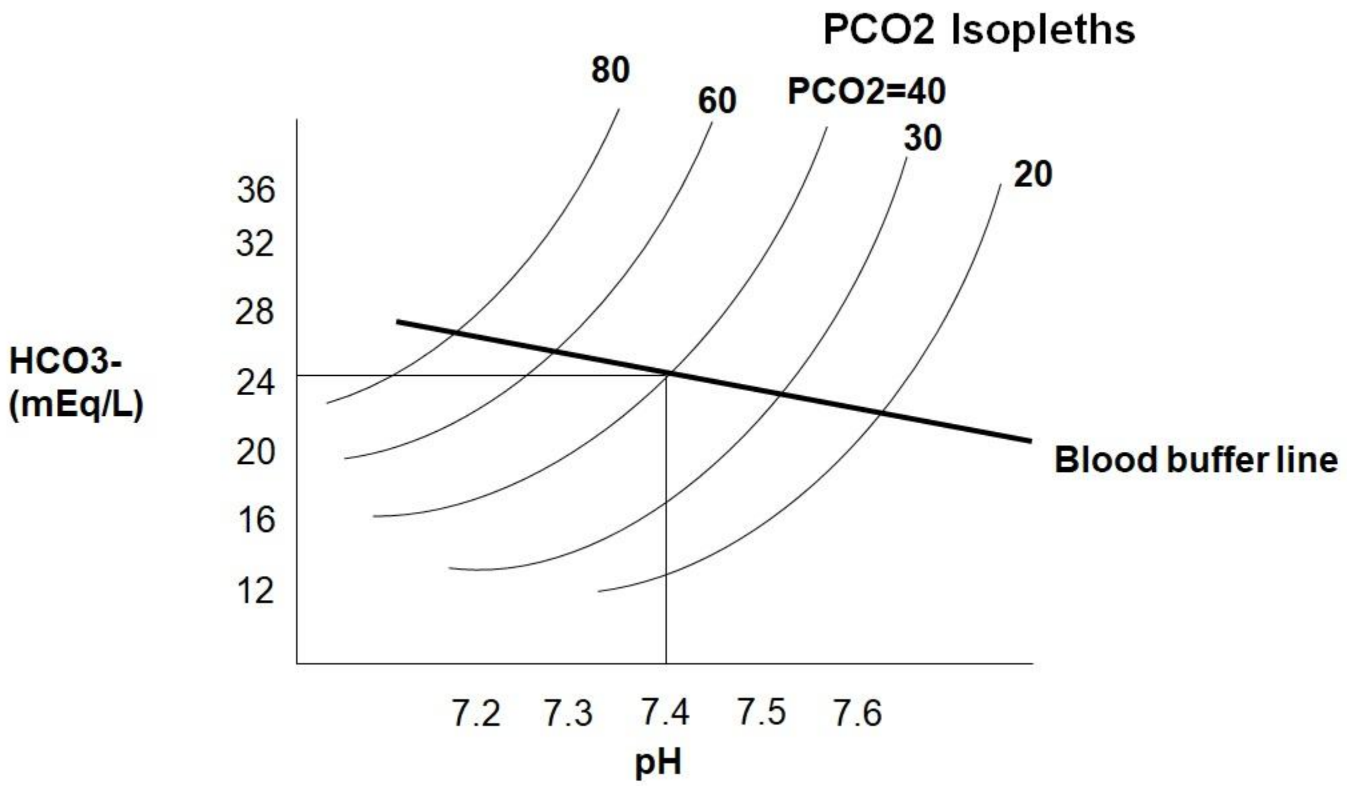
Identify the acid base disturbance;
Name a disorder;
Identify compensatory mechanisms

Davenport Diagram: Summary



Acute respiratory disorders always lie on the blood buffer line

A 14-year-old boy is brought to the physician because of a 2-day history of abdominal pain, increased urination, increased thirst and weakness. His respirations are 30/min and deep. Urinalysis shows presence of glucose and ketone bodies in urine. Blood samples are obtained for arterial blood gas analysis. Plot the most likely point on the Davenport diagram.



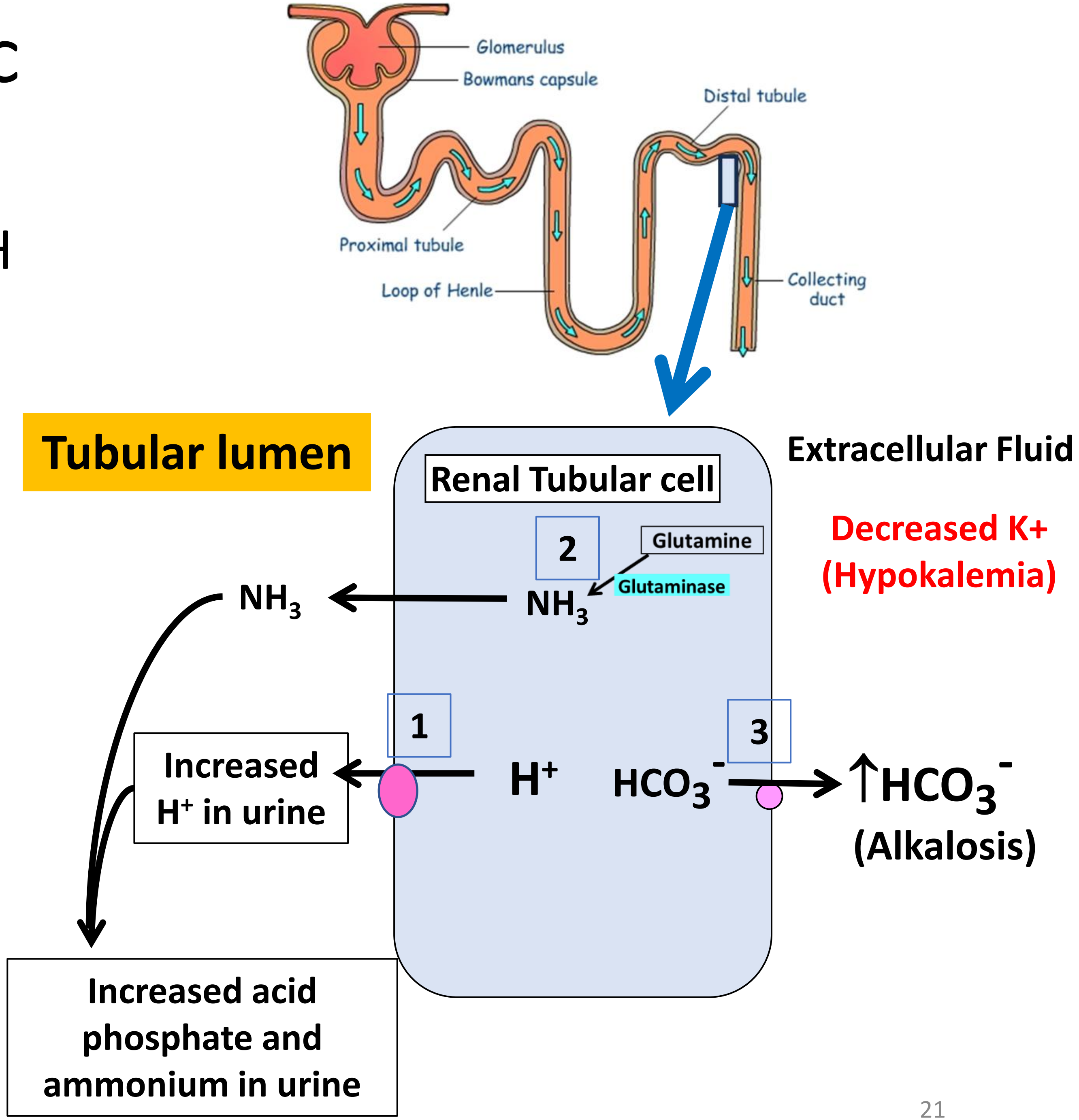
Identify the acid base disturbance and disorder;
Explain associated lab findings (Anion gap; Serum K^+);
Identify compensatory mechanisms

A 42-year-old woman comes to the physician because of a 1-week history of weakness. She says she has been using laxatives and colon cleansing agents 5 days/ week for the past 6 months for colonic health. Laboratory studies show serum potassium is 2.9 mEq/L (3.5-5 mEq/L) and arterial pH is 7.47. Which of the following additional findings is most likely?

A. Increased urinary bicarbonate	0%
B. Decreased serum bicarbonate	0%
C. Inhibition of renal carbonic anhydrase	0%
D. Increased tubular H^+ secretion	0%
E. Inhibition of renal glutaminase	0%

Hypokalemia and metabolic alkalosis (**al****K****a****LO****sis**)

- Also, changes in serum K^+ affects pH
  - Hypokalemia → Alkalosis
- Effect of Hypokalemia on **renal tubules**
  1. Stimulates H^+ secretion into lumen
  2. Stimulates ammonia formation
  3. Stimulates formation of new HCO_3^- and **increases serum HCO_3^- (Alkalosis)**
- **Hypokalemia maintains and worsens alkalosis**
- **TIP: Treat hypokalemia to resolve alkalosis**



An 18-year-old woman has the following lab reports: pH = 7.46; $PaCO_2$ = 24 mmHg; HCO_3^- = 17 mmol/L.

- What is the acid base disturbance?
- Explain why PCO_2 and HCO_3^- are abnormal.
- What compensatory mechanisms are active in this patient?

- Name 3 disorders that can result in this acid base disturbance.

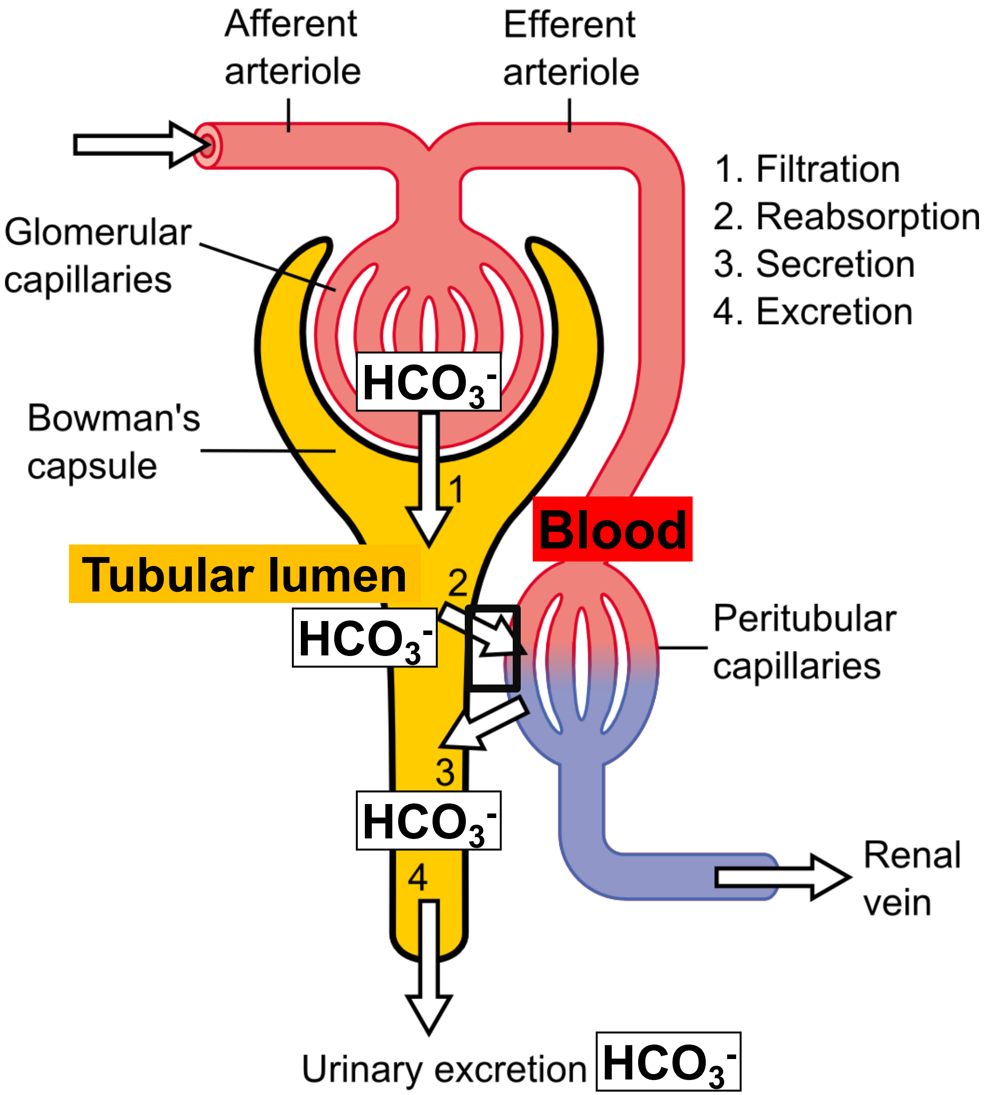
A 28-year-old woman is visiting Cusco, Peru. One week later, she has the following lab reports: pH = 7.54; $PaCO_2$ = 20 mmHg; HCO_3^- = 17mmol/L. How do the kidneys respond to this acid base disorder?
Identify adaptations in RBC and hemoglobin ODC

A. I 17%
B. II 17%
C. III 17%
D. IV 17%
E. V 17%
F. VI 17%

	Urinary bicarbonate	**Urinary acid phosphate**	**Urinary ammonium**	**Urine pH**
I	Increased	Increased	Increased	Low
II	Decreased	Decreased	Decreased	Low
III	Increased	Decreased	Decreased	Low
IV	Increased	Decreased	Increased	High
V	Decreased	Decreased	Decreased	High
VI				

Review

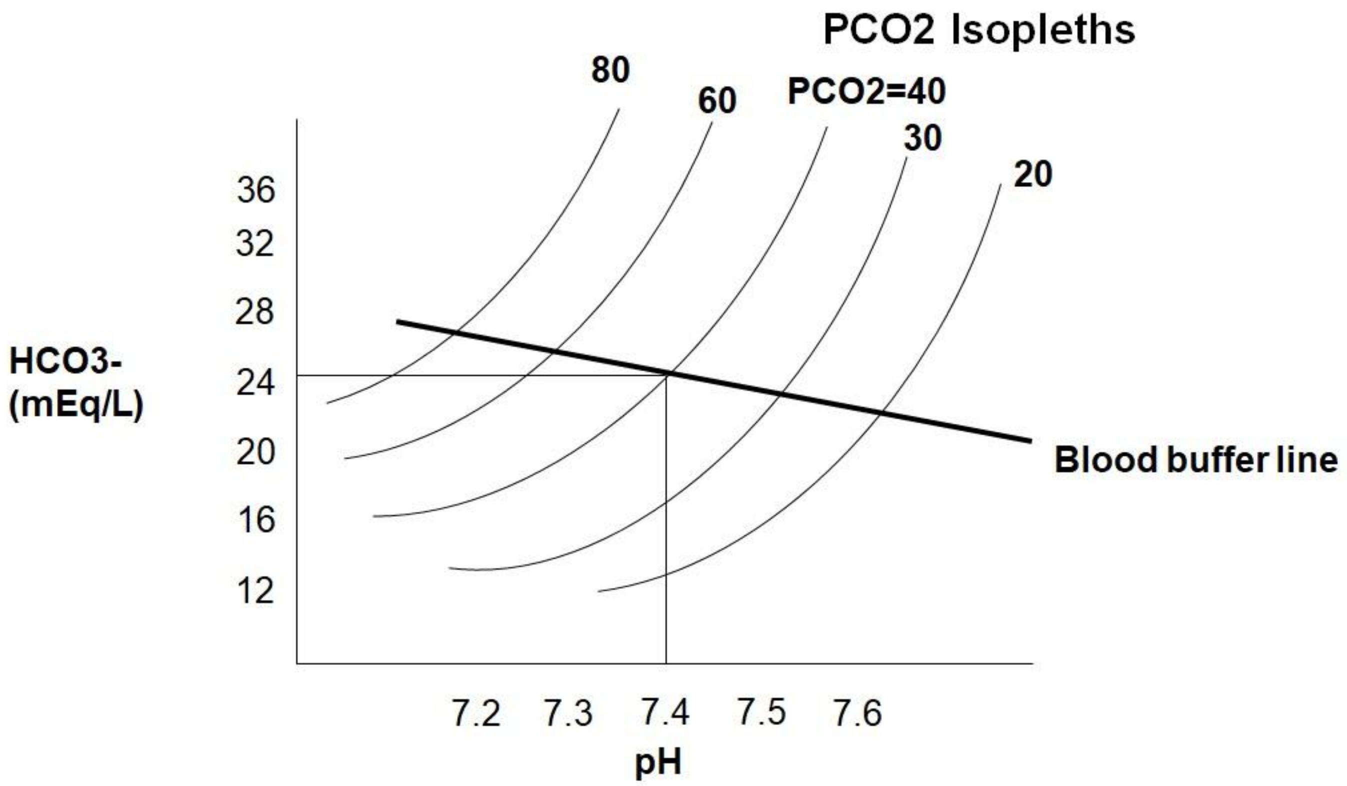
Renal compensation in respiratory/ metabolic alkalosis



Increased bicarbonate loss in urine

- In alkalosis (if kidney is normal and fluid volume is normal)
- Decreased H^+ secretion by tubule
- Less reabsorption of filtered HCO_3^-
- HCO_3^- secreted by beta-intercalated cells
- HCO_3^- loss in urine increases
- Acid phosphate and ammonium in urine decreases
- Urine pH becomes alkaline

Plot on the Davenport diagram: pH =7.17; PCO_2 = 55mmHg; HCO_3^- = 20mEq/L

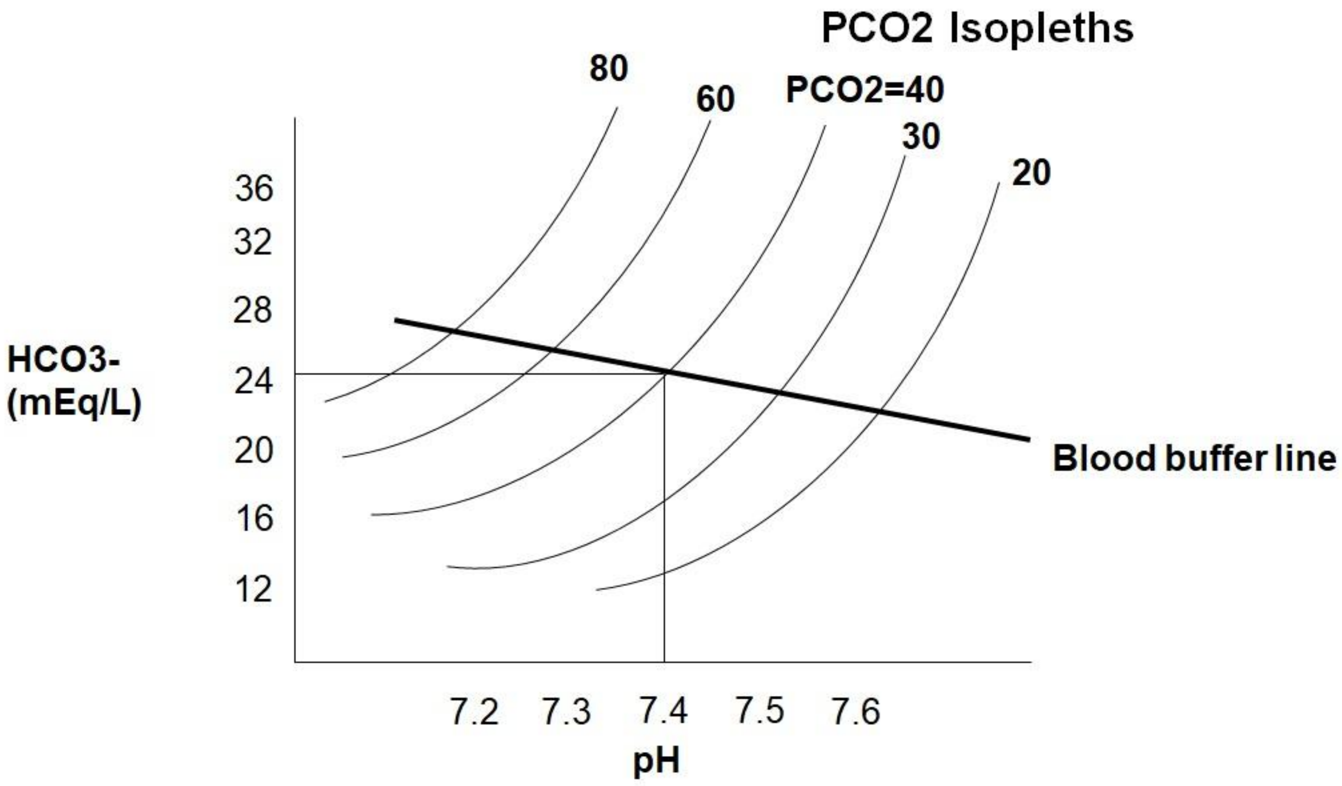


Identify the acid base disturbance

A 3-week-old boy has the following lab reports: pH = 7.48; $PaCO_2$ = 48mmHg; HCO_3^- = 35mmol/L. His pulses are weak and blood pressure is low. Skin turgor is lost.

- What is the acid base disturbance?
- Explain why $PaCO_2$ and HCO_3^- are abnormal.
- What compensatory mechanisms are most likely active?
- He is excreting an acidic urine. Why?
- Name 3 disorders that can result in this acid base disturbance

A 15-year-old boy is brought to the emergency department. ABG results show pH = 7.53; PCO_2 = 28 mmHg; HCO_3^- = 23mmol/L. He also has cramps and Trousseau sign is positive.



Identify the acid base disturbance;
Name a disorder;
Identify compensatory mechanisms

Case

- A 76-year-old man has severe sub-sternal chest pain that radiates down the inside of his left arm. Shortly afterward, he collapsed. He is unresponsive, not breathing, and without a pulse. CPR and electroconvulsive shock are required to start his heart beating again. On arrival to the Emergency Room, he regained consciousness, and has severe shortness of breath (dyspnea) and chest pain.
- His breathing is labored, pulses are rapid and weak. His skin is cold and clammy.

Blood pressure	85/50 mmHg
Heart rate	175/minute
Respiratory rate	32/minute
Temperature	99.2^{o}F

- ECG shows significant "Q" waves in most of the leads. Laboratory studies show markedly elevated cardiac **Troponin I**.

ABG on arrival: pH: 7.29; PCO_2:25mmHg; HCO_3^-:12 mEq/L; Na^+:140; Cl^-:102. What is the anion gap?

A. 6	20%
B. 10	20%
C. 16	20%
D. 26	20%
E. 30	20%

ABG on arrival: pH: 7.29; PCO_2:25mmHg; HCO_3^-:12 mEq/L; Na^+:140; Cl^-:102. Which of the following best describes this acid base disorder?

A. Metabolic acidosis	13%
B. High anion gap metabolic acidosis	13%
C. Normal anion gap metabolic acidosis	13%
D. Acute respiratory acidosis	13%
E. Chronic respiratory acidosis	13%
F. Metabolic alkalosis	13%
G. Acute respiratory alkalosis	13%
H. Chronic respiratory alkalosis	13%

Case

- What is the cause of this acid base disorder? Which acid is most likely accumulating in the patient?

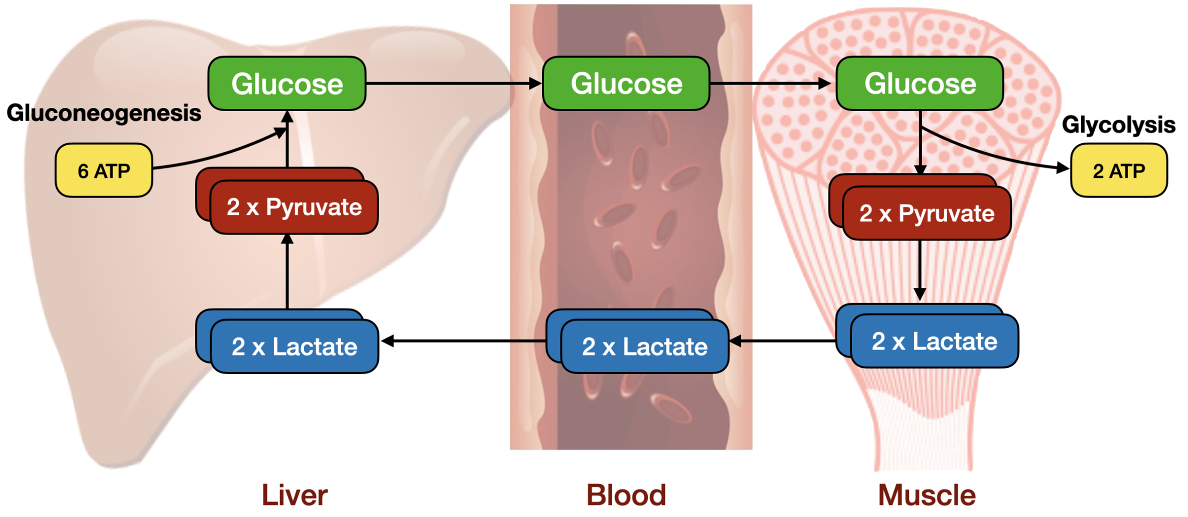
- Why is bicarbonate level low in the patient?

Why are lactate levels elevated in the patient?

A. Circulatory shock 0%

B. Increased anaerobic metabolism 0%

C. Impaired Cori cycle 0%



The ABG analysis of a 22-year-old man with a 10-year history of type-1 diabetes mellitus show: pH=7.28; $PaCO_2$=32mmHg; HCO_3^- =15mEq/L. Urine glucose and ketone bodies are positive. What additional finding is most likely in this patient?

A. Tubular H^+ secretion increases 0%
B. Anion gap is normal 0%
C. Urinary acid phosphate is decreased 0%
D. Decreased rate of breathing 0%
E. Urinary ammonium excretion decreases 0%

A patient with acute bronchitis is treated with antibiotics. No improvement is observed (antibiotic resistant respiratory infection). What is the predicted arterial blood gas results 5 days later?

A. pH-7.54; PCO_2-40; HCO_3^--34 20%
B. pH-7.3; PCO_2-70; HCO_3^--34 20%
C. pH-7.3; PCO_2-31; HCO_3^--15 20%
D. pH-7.51; PCO_2-24; HCO_3^--19 20%
E. pH-7.09; PCO_2-60; HCO_3^--18 20%

A 25-year-old woman has the following ABG results: pH: 7.5; $PaCO_2$: 30 mmHg; PO_2: 100 mmHg; Hb-O_2 saturation: 98%; HCO_3^-: 23 mEq/L. Which values are likely in the chronic state for this patient?

A. I 0%

B. II 0%

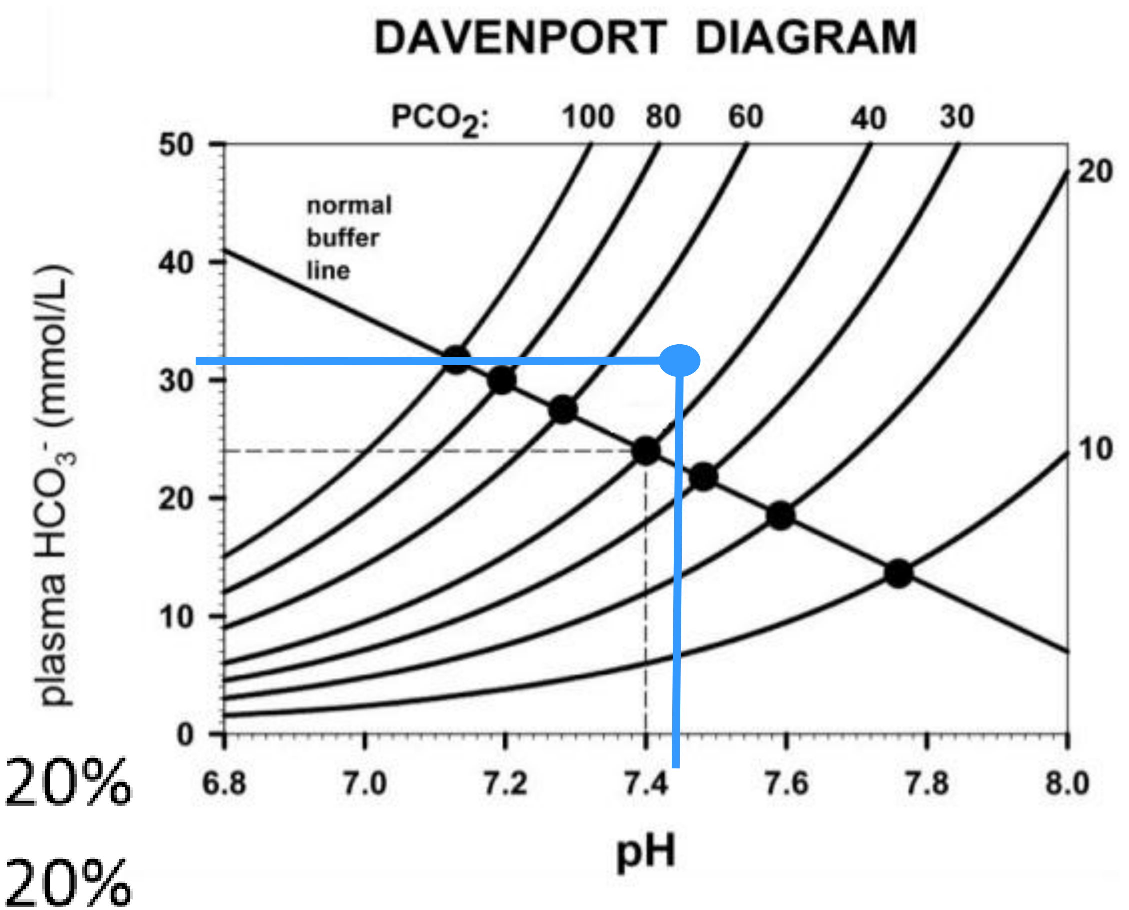
C. III 0%

D. IV 0%

E. V 0%

	pH	**HCO_3^-**	**PCO_2**
I	7.31	30	60
II	7.46	21	30
III	7.21	12	30
IV	7.47	36	50
V	7.53	29	35

The arterial blood gas findings are shown in the Davenport diagram. Which of the following conditions is most likely in this patient?



A. Chronic renal failure 20%
B. High altitude adaptation 20%
C. Panic attack causing hyperventilation 20%
D. Chronic obstructive pulmonary disease 20%
E. Hyperaldosteronism 20%

Explain additional findings in the patient

ABG: pH: 7.5; $PaCO_2$: 48 mmHg; [HCO_3^-]: 37 mmol/L. Assume fluid volume is normal. What compensatory mechanisms are active? (>1 answer)

A. Hyperventilation	0%
B. Inhibition of respiratory center	0%
C. Increased tubular H^+ secretion	0%
D. Increased HCO_3^- loss in urine	0%
E. Increased urinary titratable acid	0%
F. Increased urinary ammonium	0%

The arterial blood gas analysis of a 55-year-old woman who is a chronic smoker show: pH=7.16; PCO_2=70mmHg; HCO_3^- =25mEq/L. What disorder is most likely in this patient?

A. Diabetic ketoacidosis 0%
B. Acute respiratory distress syndrome 0%
C. Chronic renal failure 0%
D. High altitude sickness 0%
E. Pyloric stenosis 0%

The ABG analysis of a 50-year-old man show:
pH=7.24; PCO_2=80mmHg; HCO_3^-=34mEq/L. What compensatory mechanism is most likely active in this patient?

A. Bicarbonate is secreted in urine 0%
B. Urine pH becomes alkaline 0%
C. Urinary hydrogen ion secretion decreases 0%
D. Renal tubular carbonic anhydrase is stimulated 0%
E. Tubular glutaminase activity is decreased 0%

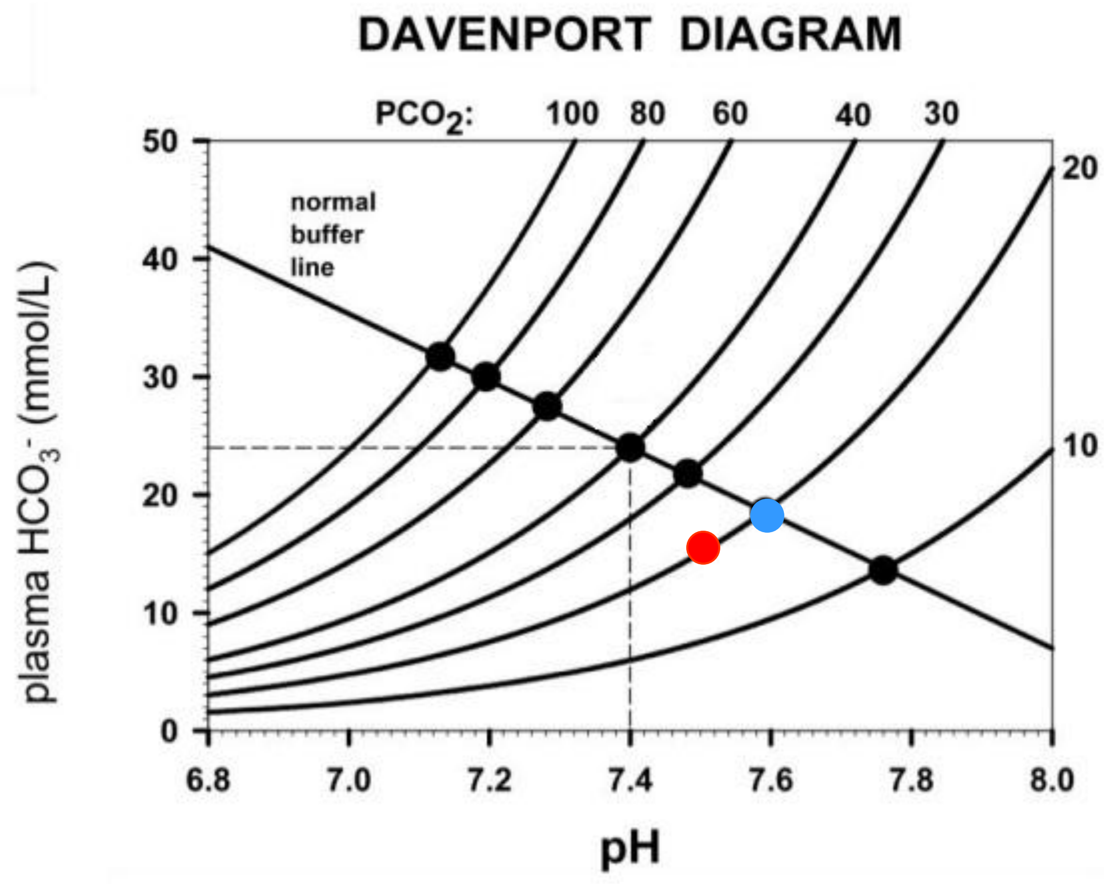
Which of the following arterial blood gas reports corresponds to the listed acid base disorder?

	pH	HCO_3^-	PCO_2
I	7.31	30	60
II	7.34	16	30
III	7.21	12	30
IV	7.47	35	49
V	7.54	30	35

A. I - Acute respiratory acidosis 0%
B. II – Chronic respiratory acidosis 0%
C. III – Metabolic acidosis 0%
D. IV – Chronic respiratory alkalosis 0%
E. V – Mixed acidosis 0%

Which of the following acid-base disorders does the red dot most likely represent?

A. Acute respiratory acidosis 14%
B. Acute respiratory alkalosis 14%
C. Metabolic acidosis 14%
D. Metabolic alkalosis 14%
E. Chronic respiratory acidosis 14%
F. Chronic respiratory alkalosis 14%
G. Mixed alkalosis 14%



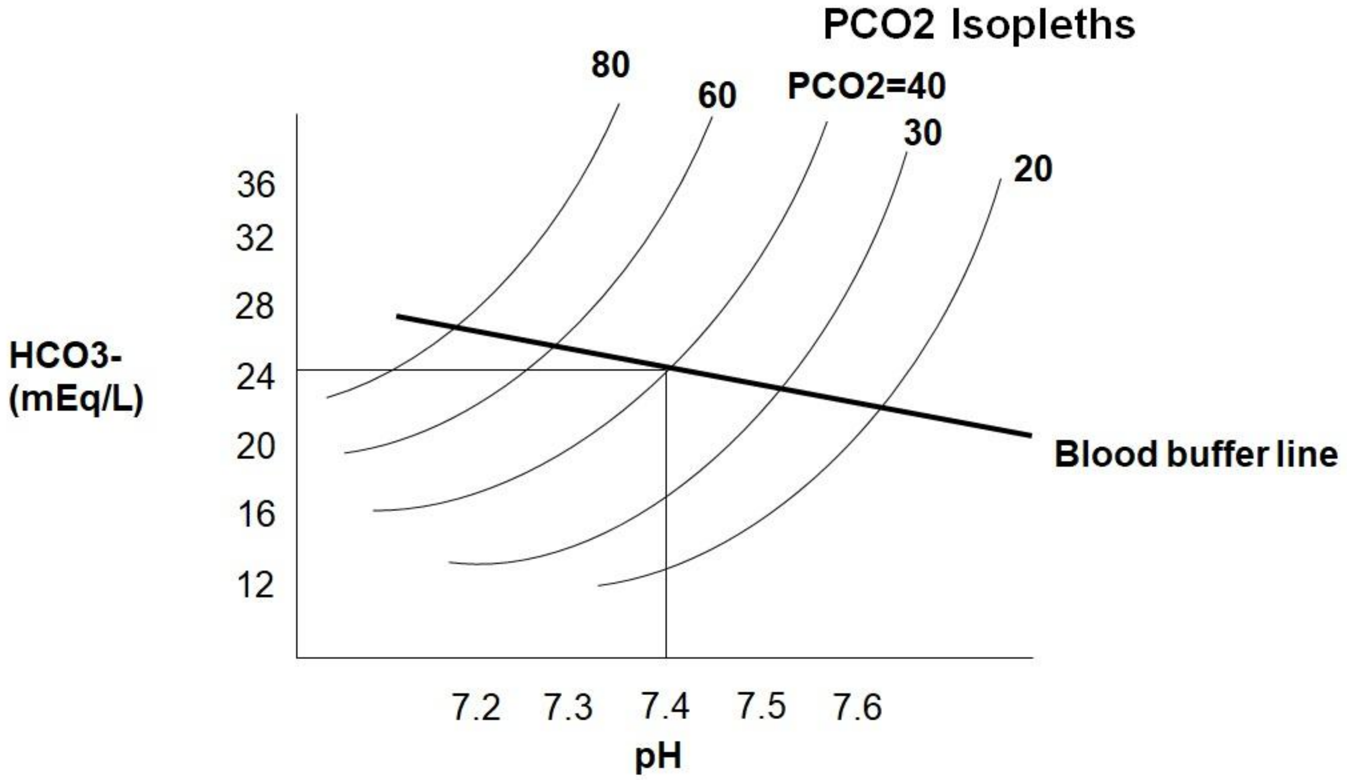
A 4-year-old boy has the following ABG reports: pH: 7.28; $PaCO_2$: 32 mmHg; $[HCO_3^-]$: 15 mmol/L. What compensatory mechanisms are most likely active?

A. Urinary HCO_3^- excretion is increased 0%
B. Respiratory center is stimulated 0%
C. Urine pH becomes alkaline 0%
D. Tubular H^+ secretion is decreased 0%
E. Respiratory rate is decreased 0%

A 14-year-old girl with cystic fibrosis has fever and increased productive cough over the past 24 hours. The sputum is green in color. She is also increasingly short of breath and is noticeably wheezing on physical examination. pH: 7.23; PCO_2: 60 mmHg; PO_2: 55 mm Hg; $[HCO_3^-]$: 25 mmol/L. What is the acid base disorder?

- Explain how cystic fibrosis causes this acid-base disturbance?

Plot the acid base status after 5 days (no response to therapy)



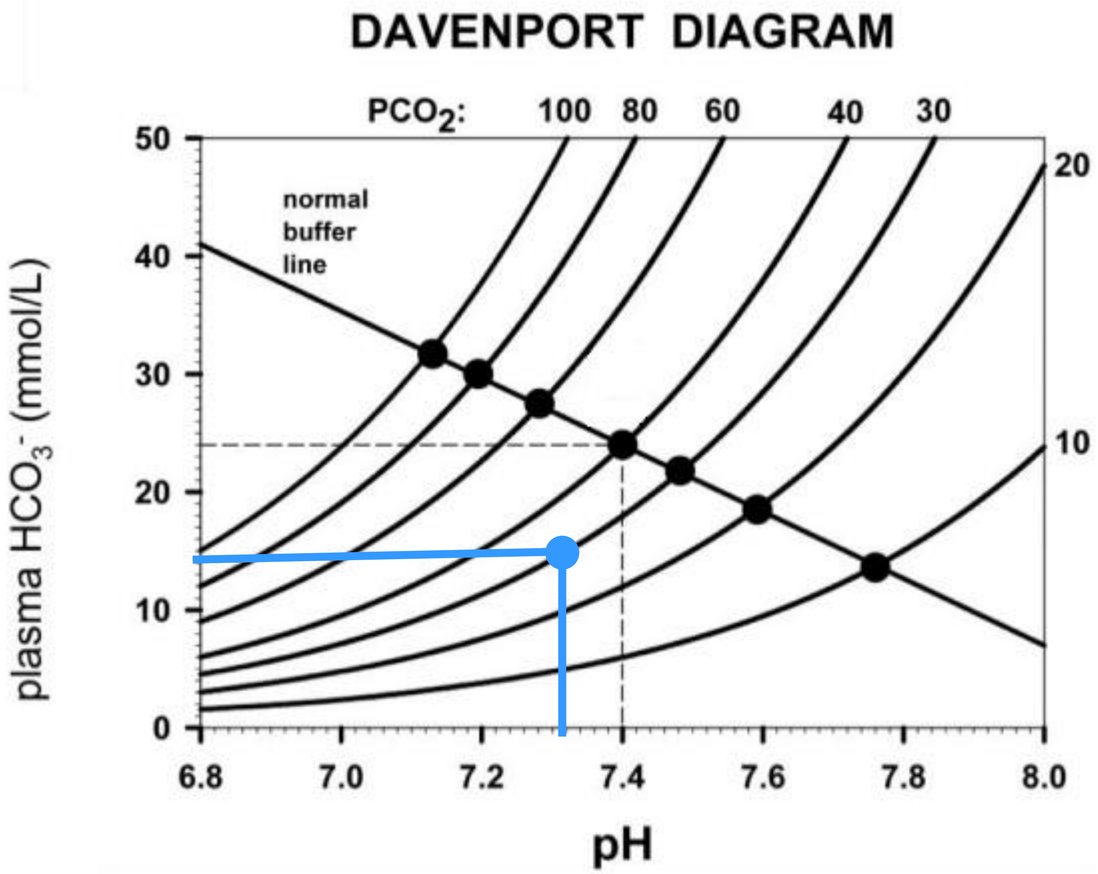
An 80-year-old man is on a ventilator in ICU. ABG: pH: 7.5; $PaCO_2$:30 mmHg; PO_2:100 mmHg; Hb-O_2 saturation: 98%; HCO_3^-: 23mEq/L. What is the most likely acid base disorder?

A. Acute respiratory acidosis	0%
B. Chronic respiratory acidosis	0%
C. Acute respiratory alkalosis	0%
D. Chronic respiratory alkalosis	0%
E. Metabolic alkalosis	0%
F. Metabolic acidosis	0%
G. Mixed alkalosis	0%

- Why is the $PaCO_2$ decreased?

What does the blue dot represent?

A. Acute respiratory acidosis 0%
B. Metabolic alkalosis 0%
C. Metabolic acidosis 0%
D. Chronic respiratory alkalosis 0%
E. Chronic respiratory acidosis 0%
F. Acute respiratory alkalosis 0%



ABG: pH 7.29; PCO_2-25mmHg; HCO_3^- 12. What compensatory mechanisms are active?

A. Respiratory center is inhibited 20%
B. Decrease in respiratory rate 20%
C. Urinary titratable acid is increased 20%
D. Urinary pH is increased 20%
E. Urinary ammonium excretion is low 20%

A 12-year-old boy has the following ABG reports: pH: 7.34; $PaCO_2$: 30 mmHg; $[HCO_3^-]$: 16 mmol/L; Na^+: 135 mEq/L; Cl^-: 110 mEq/L. What is the acid base disturbance?

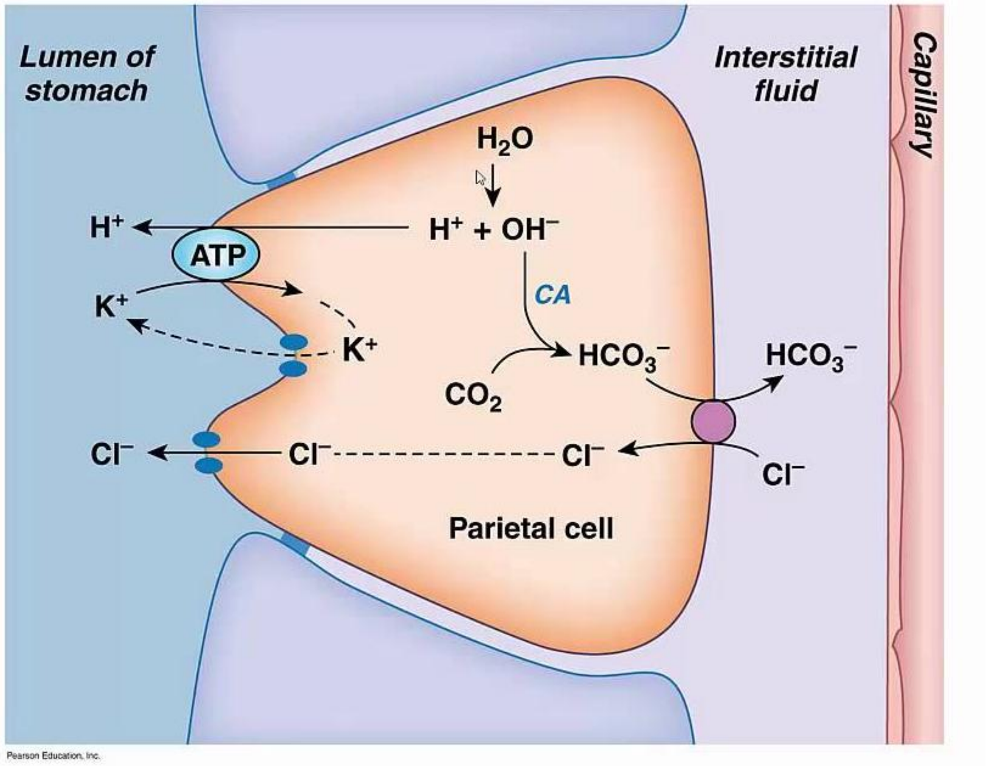
A. Metabolic alkalosis	0%
B. Chronic respiratory alkalosis	0%
C. Increased anion gap metabolic acidosis	0%
D. Acute respiratory acidosis	0%
E. Hyperchloremic metabolic acidosis	0%

A 40-year-old man has had repeated episodes of vomiting for the past two days. ABG: pH: 7.48; $PaCO_2$: 50 mmHg; $[HCO_3^-]$: 37 mmol/L. Imaging studies show pyloric obstruction due to peptic ulcer. What is the acid base disturbance?

A. Metabolic acidosis	0%
B. Acute respiratory alkalosis	0%
C. Chronic respiratory acidosis	0%
D. Metabolic alkalosis	0%
E. Acute respiratory acidosis	0%

Case: Questions

- Explain why there is an increase in arterial bicarbonate?



Case: Questions

- What compensatory mechanisms are active in this patient?

The ABG analysis of a 22-year-old woman show:
pH=7.31; $PaCO_2$ =28mmHg; HCO_3^-=14mEq/L; Na^+:138; Cl^-:110. Which of the following conditions is most likely?
Name a disorder that can cause this presentation.

A. High anion gap metabolic acidosis 0%
B. Metabolic alkalosis 0%
C. Hyperchloremic metabolic acidosis 0%
D. Chronic respiratory acidosis 0%
E. Acute respiratory alkalosis 0%
F. Mixed acidosis 0%
G. Normal anion gap acidosis 0%

The arterial blood gas analysis of a 17-year-old woman show: pH=7.51; PCO_2=30mmHg; HCO_3^-=23 mEq/L. What disorder is most likely in this patient?

A. Diabetic ketoacidosis 0%
B. Acute respiratory distress syndrome 0%
C. Chronic renal failure 0%
D. Hyperventilation due to anxiety 0%
E. Pyloric stenosis 0%
F. Chronic obstructive pulmonary disease 0%

Did you enjoy and learn in the session?

A. Yes	0%
B. No	0%
C. Maybe	0%


GOOD LUCK
FOR YOUR
EXAM AND
DO THE BEST